

# **IMU-based Speed and Distance Estimation Analysis**

## 1. Data Overview

**Sampling Rate:** 149.73 Hz  
**Total Duration:** 423.18 seconds  
**Number of Samples:** 63364  
**Nyquist Frequency:** 74.87 Hz

## 2. Signal Processing Pipeline

### 2.1 Pre-processing:

- **Butterworth Low-pass Filter** (IMU data)  
Removes high-frequency noise, preserves motion dynamics.
- **Nyquist Theorem:** Sampling rate must be at least twice the highest frequency of interest.

### 2.2 Orientation Estimation:

- **Gyroscope Integration (Euler angles):** Simple numerical integration to estimate roll, pitch, yaw.

### 2.3 Coordinate Transformation:

- **Body to World Frame:** 3D rotation matrix compensates for device orientation.

### 2.4 Acceleration Processing:

- **High-pass Filter:** Removes integration drift.

### 2.5 Visual: Raw and Filtered Accelerometer Magnitude

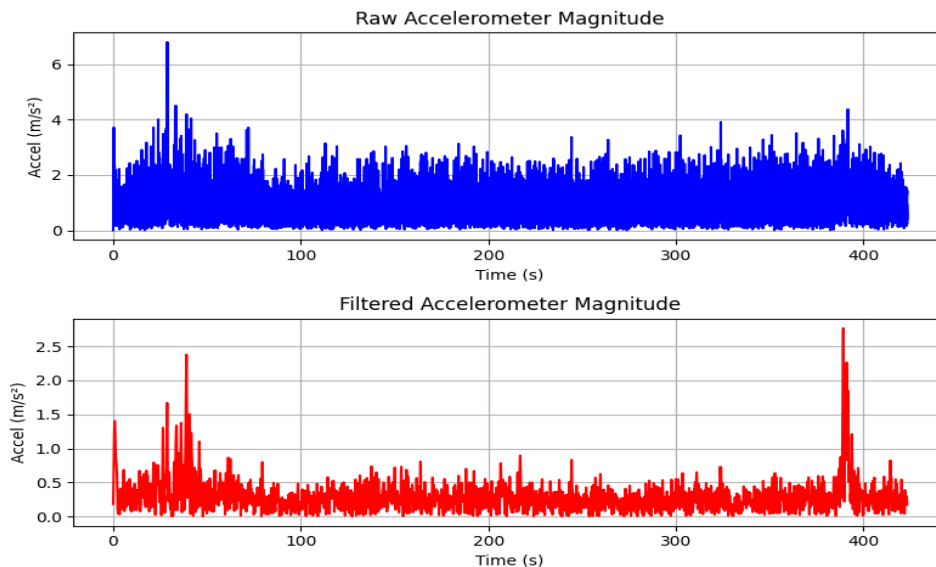


Figure: Shows the effect of low-pass filtering on the accelerometer signal. Filtering is crucial for removing noise before integration.

### 2.6 Visual: FFT Before and After Filtering (All Axes)

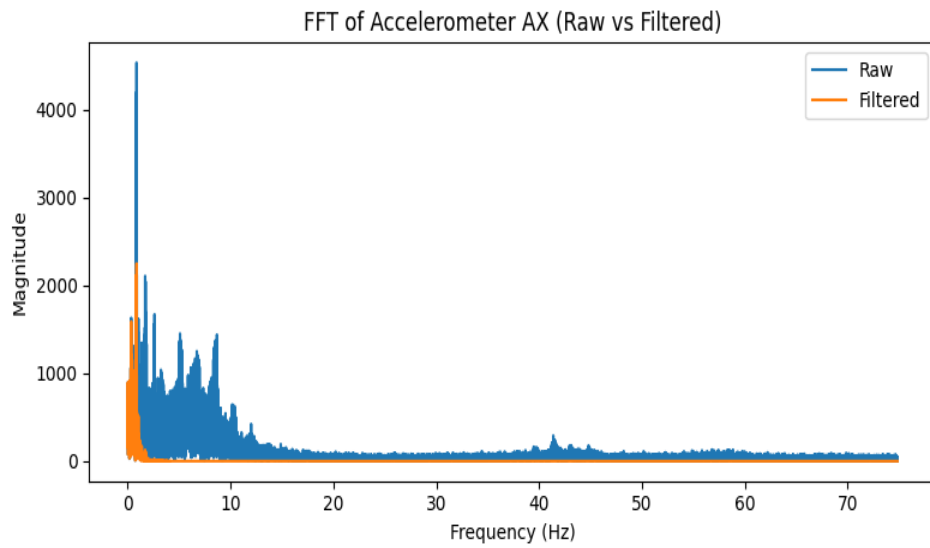


Figure: FFT Accelerometer AX (Raw vs Filtered in Frequency Domain)

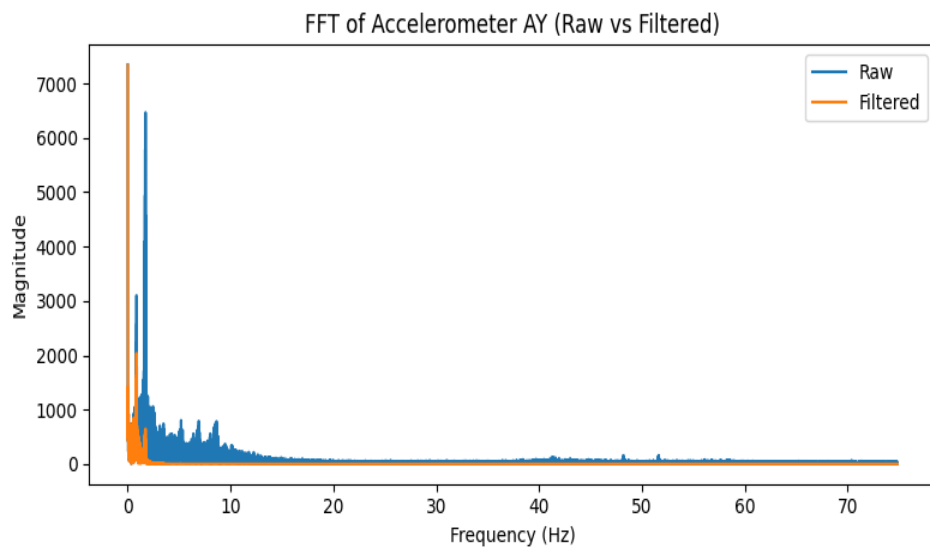


Figure: FFT Accelerometer AY (Raw vs Filtered in Frequency Domain)

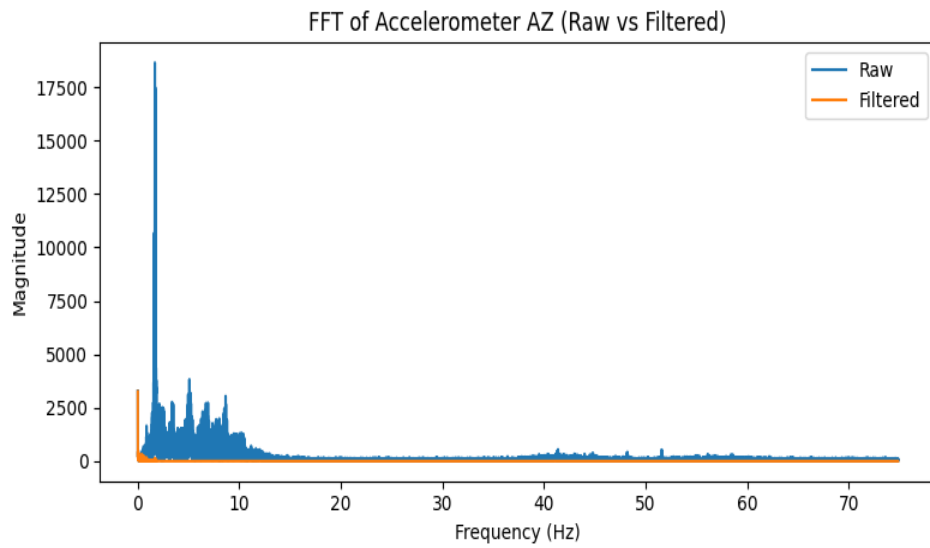


Figure: FFT Accelerometer AZ (Raw vs Filtered in Frequency Domain)

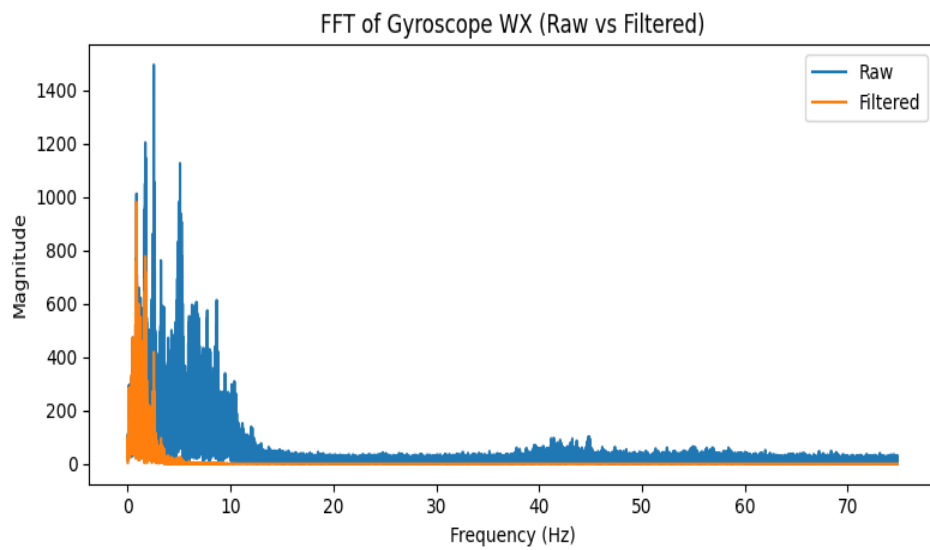


Figure: FFT Gyroscope WX (Raw vs Filtered in Frequency Domain)

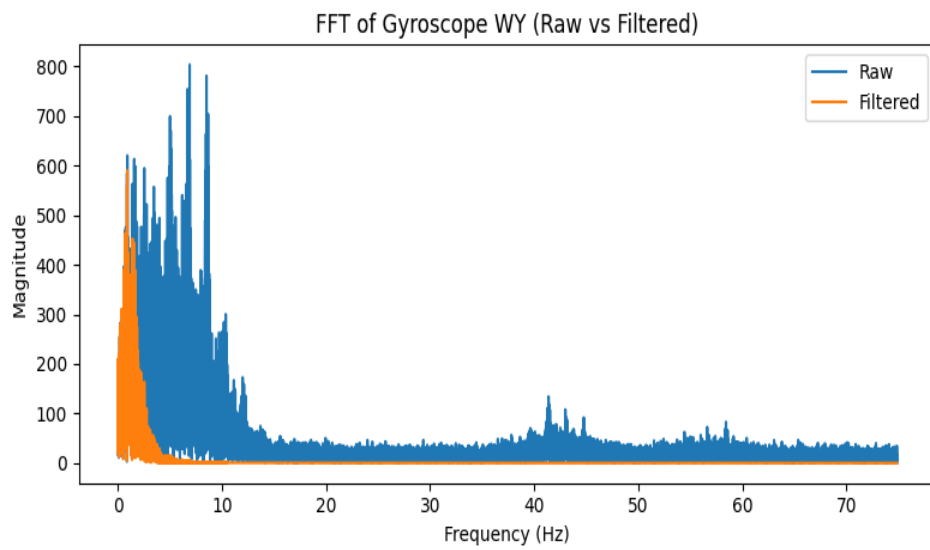


Figure: FFT Gyroscope WY (Raw vs Filtered in Frequency Domain)

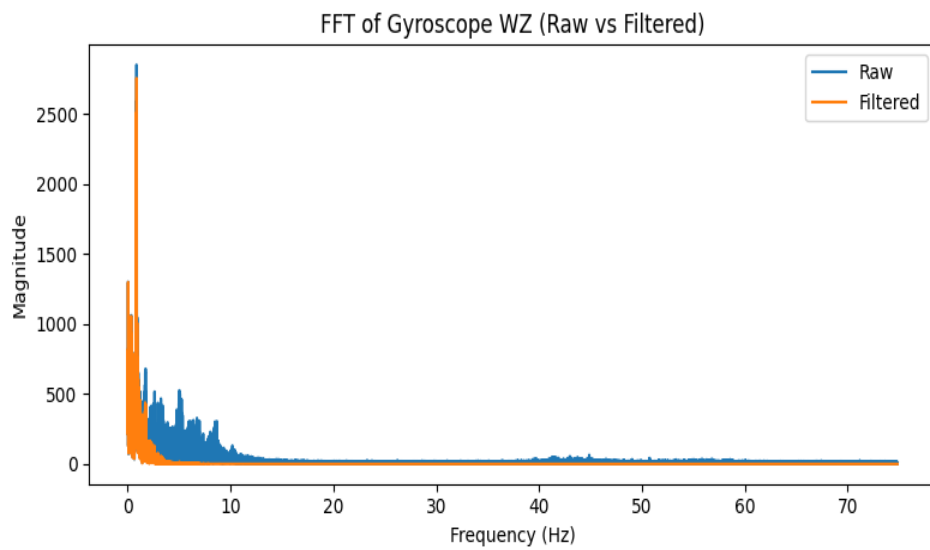


Figure: FFT Gyroscope WZ (Raw vs Filtered in Frequency Domain)

### 3. Machine Learning Analysis

#### 3.1 Feature Extraction:

- Window Size: 1.0 seconds
- Features: Mean, Std, Max, Min, Peak-to-peak of IMU signals

**3.2 Model:** Random Forest Regressor

**3.3 Performance:** RMSE = 0.089 m/s

#### 3.4 Visual: Speed Estimation Comparison

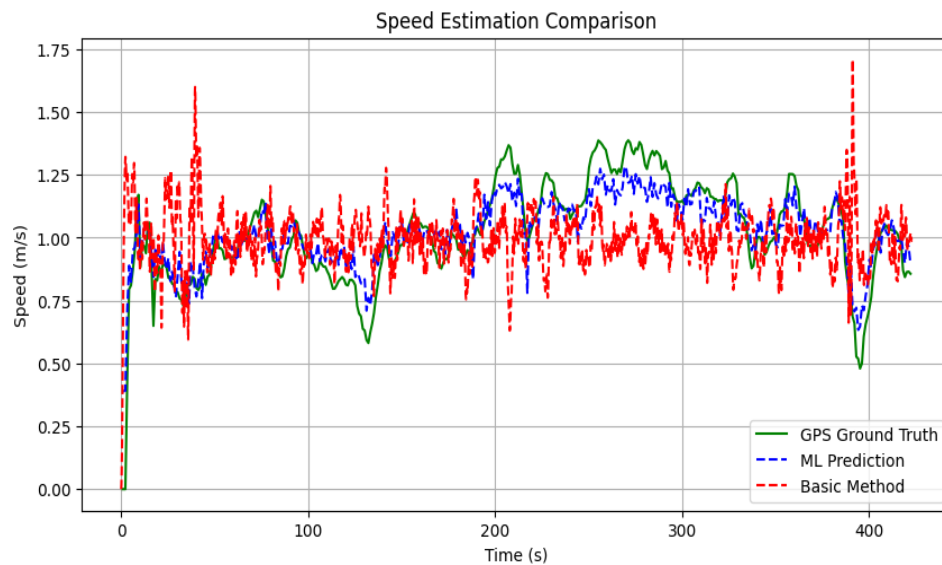


Figure: Comparison of speed estimated by GPS, ML, and basic integration. ML can learn to correct for drift and noise.

## 4. Methods Comparison

| Method             | Total Distance (m) | Relative Error (%) |
|--------------------|--------------------|--------------------|
| GPS (Ground Truth) | 432.13             | 0.00               |
| ML Method          | 431.20             | 0.22               |
| Basic Method       | 421.14             | 2.54               |

### 4.1 Visual: 2D Path Comparison (GPS, Basic, ML)

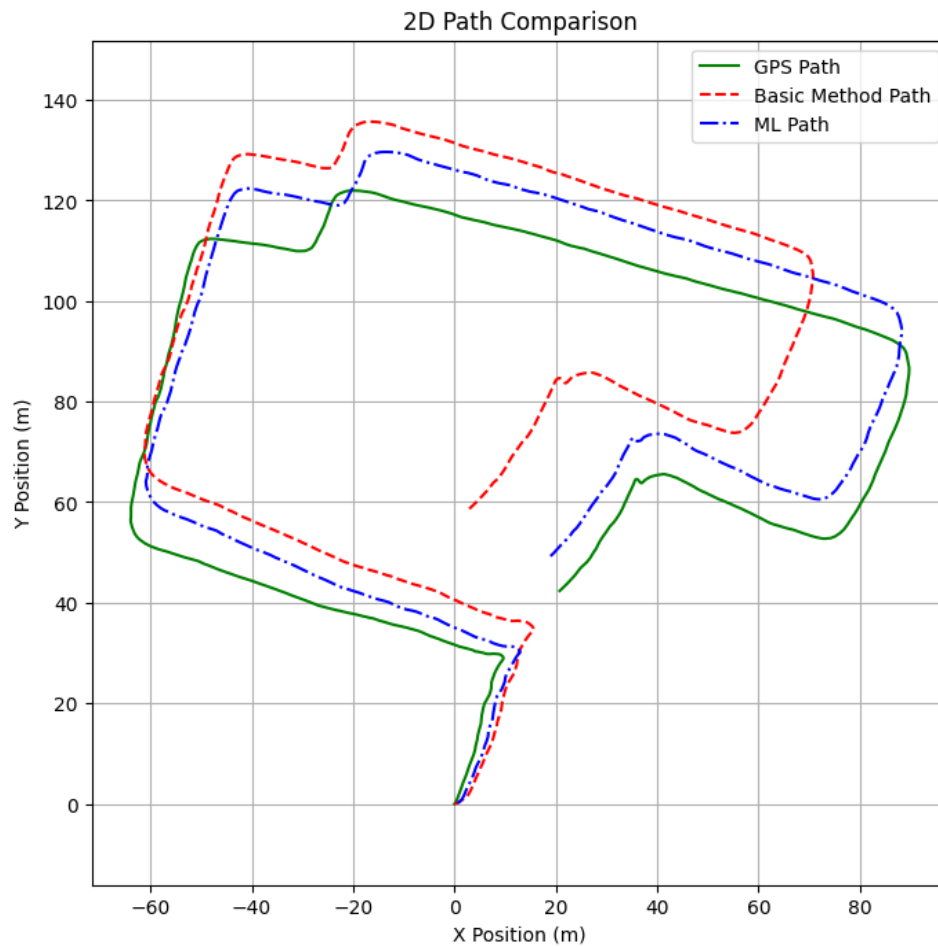


Figure: Comparison of reconstructed paths using GPS, basic integration, and ML-predicted velocities. ML can reduce drift and error compared to basic integration.

## 5. Error Analysis & Signal Processing Concepts

### 5.1 Sources of Error:

- **Integration Drift:** Small errors in acceleration accumulate over time.
- **Sensor Noise:** IMU noise is filtered using Butterworth filters.
- **Orientation Errors:** Gyroscope drift affects coordinate transformation.

### 5.2 Signal Processing Concepts:

- **Filtering:** Butterworth filters are used for both low-pass (noise removal) and high-pass (drift removal).
- **Sampling Theorem:** Ensures no aliasing in digital signals.
- **Numerical Integration:** Trapezoidal rule is used for velocity and position estimation.
- **Feature Engineering:** Statistical features from IMU signals are used for ML.

## 6. Additional Visualizations

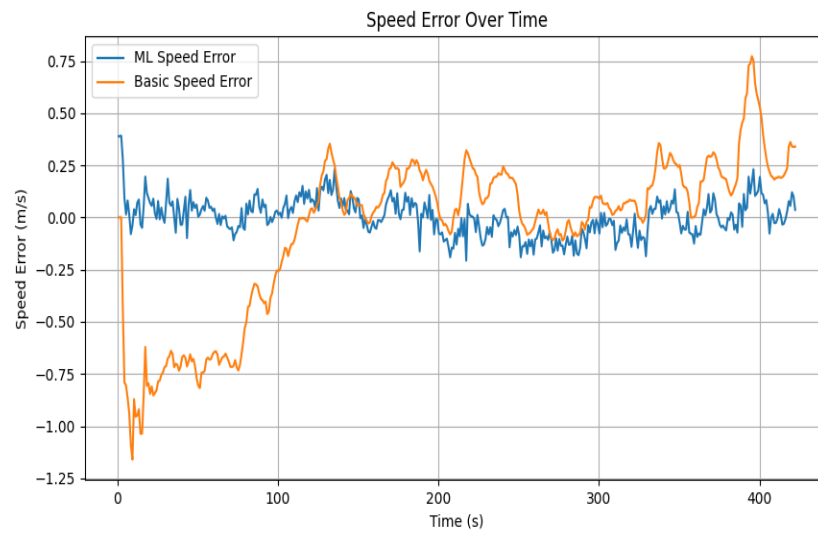


Figure: Speed Error Over Time (ML & Basic)

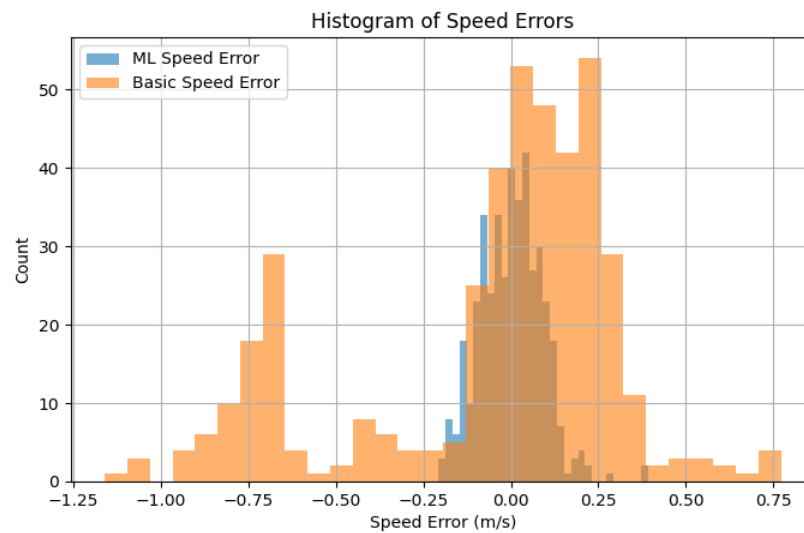


Figure: Histogram of Speed Errors

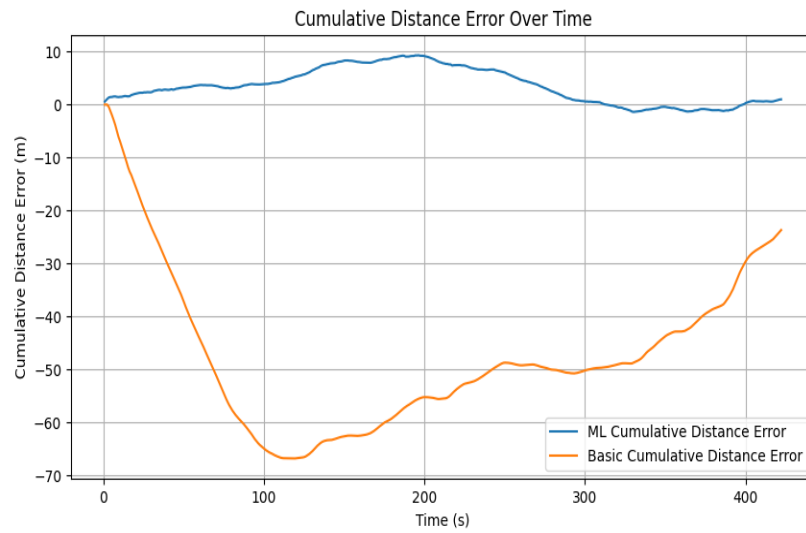


Figure: Cumulative Distance Error Over Time

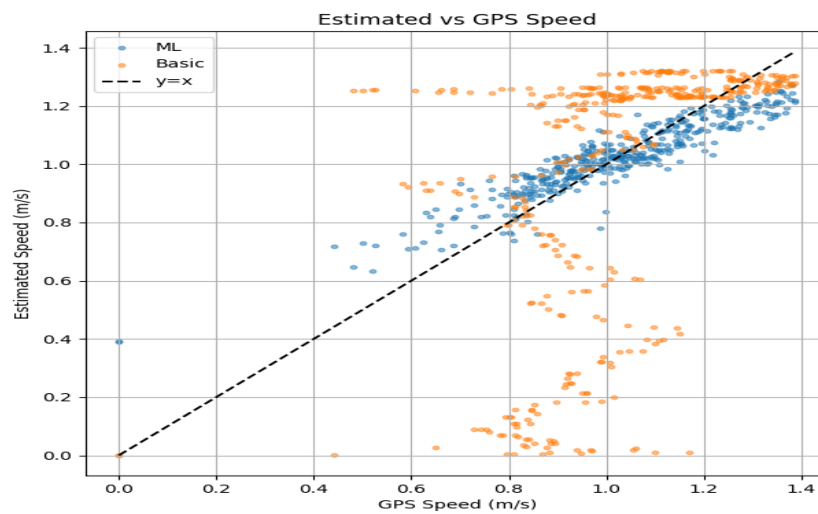


Figure: Scatter: Estimated vs GPS Speed

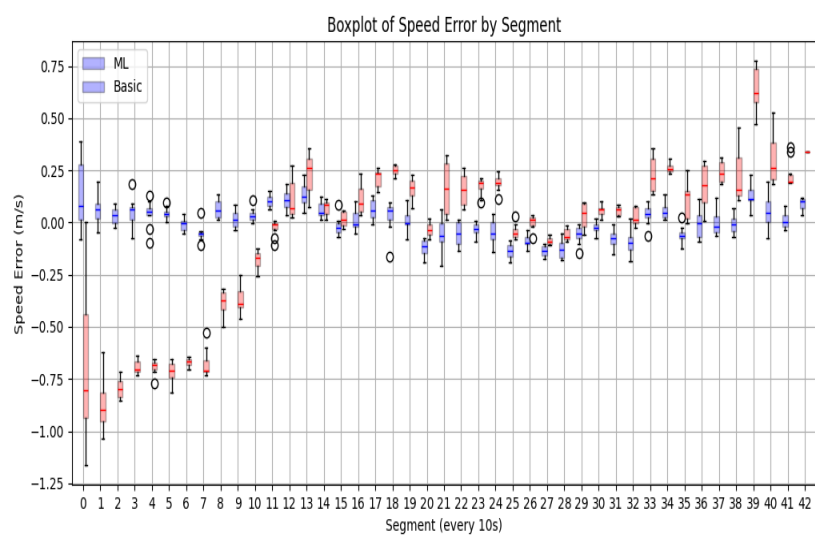


Figure: Boxplot of Speed Error by Segment

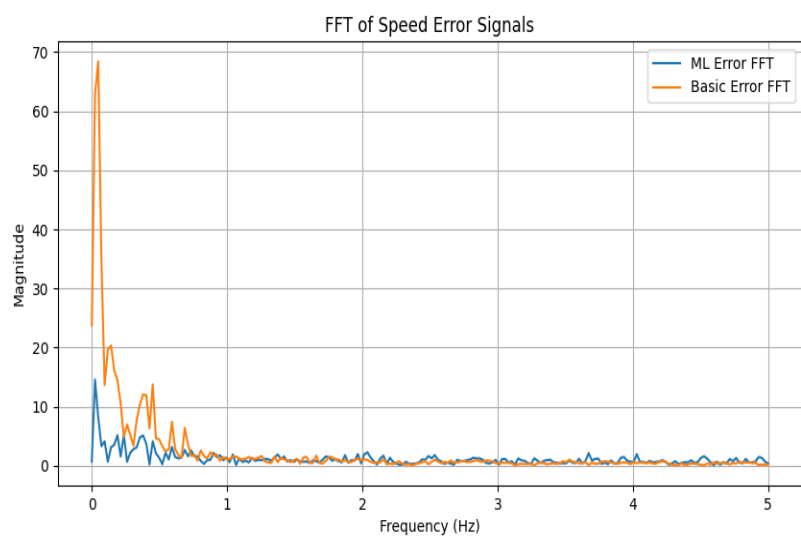


Figure: FFT of Speed Error Signals

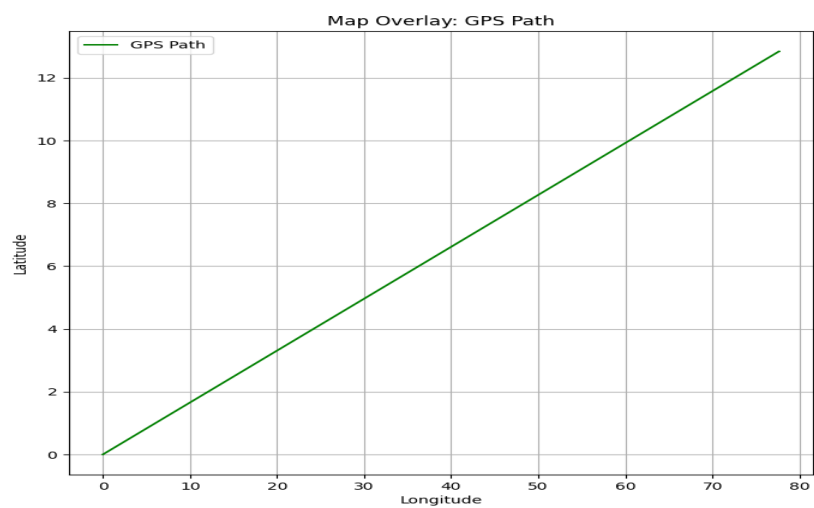


Figure: Map Overlay: GPS Path

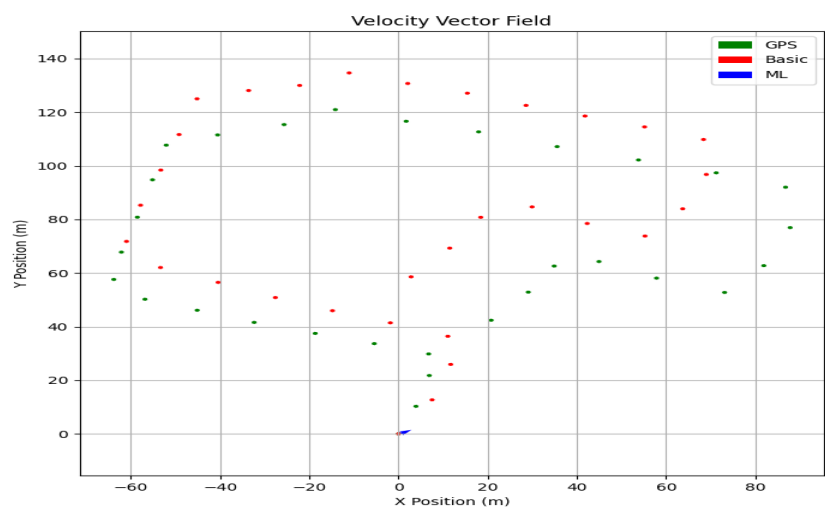


Figure: Velocity Vector Field

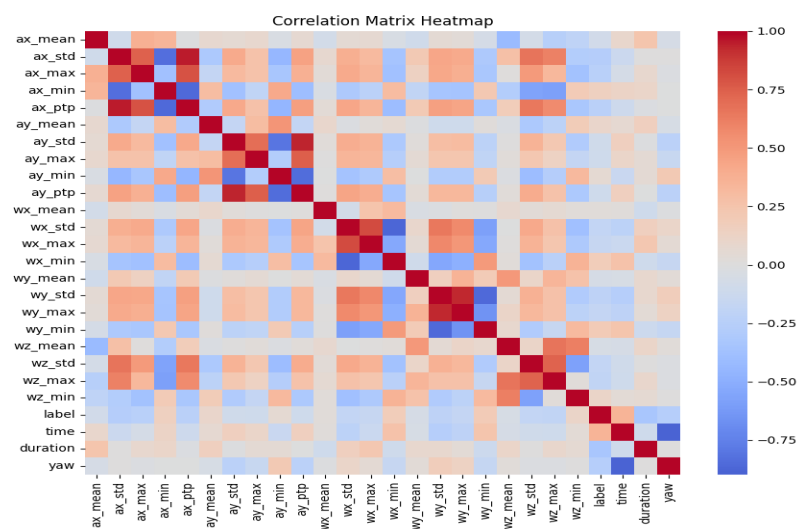


Figure: Correlation Matrix Heatmap